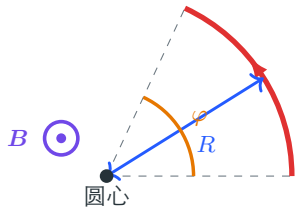
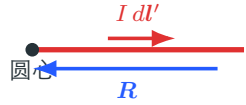
圆弧中心场：圆弧有贡献，径向导线贡献为零

半径 R、圆心角 φ 的电流弧



$\boldsymbol{B}_O = \dfrac{\mu_0 I\varphi}{4\pi R}\boldsymbol{n}$，其中 φ 必须用弧度，$\boldsymbol{n}$ 由右手定则确定

径向引线为何不产生圆心磁场



$$d\boldsymbol{l}' \parallel \boldsymbol{R} \quad \Longrightarrow \quad d\boldsymbol{l}' \times \boldsymbol{R} = \boldsymbol{0}$$

半圆弧加两条径向引线：$B_O = \dfrac{\mu_0 I}{4R}$